ECS736P/U – Information Retrieval

Assignment 1: Review of Related Work and Search Engine Design

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Abstract

This report presents the design of a multi-stage hybrid search engine for financial passage retrieval over the FiQA corpus. The central challenge is vocabulary mismatch: users ask questions in everyday language, but the corpus is written in technical financial terminology. We investigate two strategies for tackling this at the first stage of retrieval hybrid BM25 and dense bi-encoder fusion via Reciprocal Rank Fusion (Pipeline A), and Bo1 pseudo-relevance feedback query expansion (Pipeline B) both followed by a BERT cross-encoder reranker. The central experimental question is which first-stage strategy neural or enriched lexical more effectively raises recall in a domain where query and document vocabulary systematically diverge. The report covers the motivating literature, system design, evaluation strategy, and project workplan.

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1 Introduction

Cascade retrieval pipelines, in which a fast first-stage retriever generates candidates for a slower neural reranker, are now standard in information retrieval. In this architecture, first-stage quality is decisive: the reranker can only reorder documents that the first stage has retrieved, making first-stage recall a hard ceiling on end-to-end performance. A relevant document missed at Stage 1 is irrecoverable, regardless of the reranker's performance.

When the first stage is lexical, this ceiling is governed by vocabulary mismatch: the retriever cannot match a query to a document unless they share surface-form terms. Thakur et al. [17] demonstrated that this limitation is particularly damaging in specialised domains: across the BEIR benchmark, BM25 baselines on domain-specific collections such as FiQA (finance), SciFact (biomedical), and NFCorpus (nutrition) fall substantially below those on general-domain corpora, confirming that vocabulary mismatch is not a marginal issue but a primary bottleneck for domain-specific retrieval.

To address this problem, in this work, we will implement and compare two families of approaches in order to raise this ceiling. The first augments the retrieval system itself by adding a dense semantic retriever alongside BM25 and fusing their outputs. The second augments the query by enriching it with related terms before lexical retrieval. Both address the same bottleneck, but differ in architectural complexity, computational cost, and failure modes.

2 Review of Related Work

2.1 Classical Lexical Retrieval

The foundation of information retrieval (IR) was established by Salton and Buckley [1], who provided a comprehensive experimental comparison of term-weighting schemes in the vector space model. Their TF-IDF formulation assigns each term a weight proportional to its frequency in a document (TF) and inversely proportional to its frequency across the corpus (IDF), reflecting the intuition that terms appearing in few documents are more discriminative. While conceptually elegant, the original TF-IDF formulation penalises neither excessive term repetition nor document length properties that harm ranking quality in practice.

Robertson et al. [3] addressed these weaknesses in their submission to TREC-3, introducing what became known as BM25 (Best Match 25). BM25 addresses both problems through two parameters. The saturation parameter k_1 makes term frequency subject to diminishing returns: the score gain from a term's first occurrence is large, but each subsequent occurrence contributes progressively less, asymptotically approaching a ceiling a document mentioning "inflation" fifty times is not fifty times more relevant than one mentioning it twice. The length normalisation parameter b controls how strongly document length affects scoring: as $|d|$ grows relative to the corpus average, the effective contribution of term frequency is downweighted, preventing long documents from being systematically favoured. The resulting formula substantially outperformed TF-IDF on TREC collections and established BM25 as the de facto standard for first-stage retrieval a position it retains over thirty years later. We adopt BM25 as Stage 1 of our pipeline, a choice directly motivated by this work.

Fang et al. [8] formalise axiomatic constraints for "reasonable" retrieval functions, including term-frequency saturation and document-length normalisation. While no representative formula satisfies all constraints unconditionally, their analysis shows that BM25 satisfies most constraints under mild conditions, and that a simple modification of its IDF satisfies all constraints, supporting BM25 as a theoretically well-justified first-stage retriever.

A parallel line of work reframed retrieval as a generative problem. Ponte and Croft [4] proposed building a word probability distribution for each document and ranking documents by the likelihood that this distribution would produce the query terms. The framework is theoretically strong, but any query term

absent from a document produces a zero probability, collapsing the document's score entirely; it is the same vocabulary dependency problem as BM25, expressed differently.

Zhai and Lafferty [5] addressed this by assigning absent terms a probability proportional to their frequency across the entire collection rather than zero: a term common in the corpus receives a small but meaningful probability, while a rare term receives almost none. This process is called "smoothing". They compared two methods: Jelinek-Mercer interpolation, which applies a fixed mixing ratio uniformly across all documents, and Dirichlet smoothing, which applies heavier smoothing to shorter documents whose term statistics are less reliable, and lighter smoothing to longer ones. This adaptive behaviour implicitly penalises length in a manner analogous to BM25's normalisation parameter. Empirically, Dirichlet smoothing consistently outperformed Jelinek-Mercer on short title queries across multiple TREC collections, while Jelinek-Mercer performed comparably or better on longer verbose queries; overall, Dirichlet remains a robust default (Zhai and Lafferty, 2004)

Despite its stronger theoretical grounding, the language modelling framework shares the same fundamental limitation as BM25: relevance is still determined by surface-form term overlap, and the vocabulary mismatch problem remains unresolved. Given that well-tuned BM25 achieves empirically comparable performance across diverse benchmarks [17] while offering simpler implementation and native support in standard IR toolkits, we adopt BM25 as our first-stage retriever rather than a language model variant.

A critical limitation shared by all classical models is vocabulary dependency: BM25 treats terms as atomic symbols with no inherent relationship, so a document containing *equity valuation* will not be retrieved for a query about *stock pricing* unless the two phrases share surface-form tokens. This vocabulary mismatch problem becomes particularly acute in specialised domains with rich technical terminology, such as the financial domain including our dataset, FiQA which constitutes the primary motivation for the neural components in our system.

2.2 Neural Language Models for Retrieval

The introduction of BERT [9] represented a paradigm shift in natural language processing. BERT acquires this capability through pre-training on large unlabeled corpora (BooksCorpus: 800M words; Wikipedia: 2,500M words) it learns to predict words randomly hidden in sentences, which forces it to infer meaning from surrounding context. The result is that each token receives a representation shaped by its full sentential context rather than a fixed dictionary definition: the word *bank* is encoded differently in *river bank* and *investment bank*, and *cardiac arrest* and *heart attack* are mapped to nearby positions in the representation space despite sharing no identical tokens.

Although BERT was not designed for retrieval, Nogueira and Cho [10] were among the first to apply BERT directly to passage re-ranking, fine-tuning it on the MS MARCO dataset to produce a binary relevance score for each (query, passage) pair.

Their cross-encoder architecture, feeding the concatenated query and document through BERT and classifying the `[CLS]` token nearly doubled MRR@10 on the MS MARCO passage ranking dev set (from 18.7 to 36.5) and more than doubled MAP on TREC-CAR (from 15.6 to 33.5). The cross-encoder forms Stage 2 of our pipeline, and this work provides its direct empirical justification.

However, the cross-encoder architecture carries a fundamental computational constraint: scoring requires one full forward pass per query-document pair. This makes cross-encoder-based full-collection retrieval not feasible for a collection of one million documents, scoring would require millions of BERT forward passes per query. This constraint motivates the cascade design: BM25 performs exhaustive retrieval at millisecond latency, producing a manageable candidate set of 100 documents over which the cross-encoder operates.

2.3 Dense and Late-Interaction Retrieval

Karpukhin et al. [11] introduced Dense Passage Retrieval (DPR), a bi-encoder architecture that encodes queries and documents independently into dense vector representations, enabling retrieval via approximate nearest-neighbour search. Unlike the cross-encoder, the bi-encoder pre-computes document embeddings offline, allowing sub-millisecond retrieval at query time. DPR outperformed BM25 significantly on open-domain question answering benchmarks. However, because query and document are encoded independently, the model cannot capture fine-grained term-level interactions between them, resulting in lower precision than cross-encoders on re-ranking tasks [17]. DPR therefore represents a viable alternative first-stage retriever that addresses vocabulary mismatch, but we favour BM25 for three reasons: it requires no GPU-accelerated indexing, its recall on the FiQA collection is sufficient to serve as an effective candidate generation stage, and it constitutes the standard first-stage baseline in the BEIR benchmark, enabling direct comparison with published results.

Khattab and Zaharia [12] proposed ColBERT, a late-interaction model that represents queries and documents as sequences of token embeddings rather than single vectors. Relevance is computed via a MaxSim operator over all query-document token pairs, capturing richer interactions than DPR while preserving the pre-computation advantage of bi-encoders. ColBERTv2 [13] further improved efficiency through residual compression of document embeddings, reducing storage requirements by an order of magnitude. While late-interaction models represent a compelling middle ground between efficiency and accuracy, their deployment requires custom indexing infrastructure and their performance advantage over BM25 + cross-encoder pipelines on out-of-domain benchmarks remains inconsistent [17]. However, these architectures could solve a more fundamental problem: in any cascade pipeline, first-stage recall acts as a hard ceiling on overall performance. If BM25 fails to retrieve a relevant document, because the query and document share no surface-form tokens, no downstream reranker can recover it. This is the vocabulary mismatch bottleneck identified in Section 2.1, and it motivates the hybrid retrieval strategy discussed in the following section.

2.4 Hybrid Retrieval

A separate line of work has explored whether lexical and dense retrievers can be combined rather than treated as alternatives. The core insight is that these two families of models have complementary blind spots: BM25 cannot retrieve documents that express the same concept in different words, while dense models may miss documents whose relevance depends on exact or rare terms [11, 17]. Running both retrievers and merging their candidate lists allows each to compensate for the other's weaknesses. Cormack et al. [16] showed that Reciprocal Rank Fusion, a simple merging method that combines ranked lists based solely on document positions without requiring comparable scores, consistently outperformed any individual retrieval system, including supervised approaches. The fused candidate set achieves higher recall than either retriever alone, directly addressing the first-stage ceiling identified in the previous section.

For such a hybrid to be computationally feasible at first-stage scale, the dense component must support offline pre-computation of document embeddings. Reimers and Gurevych [15] demonstrated that BERT fine-tuned in a Siamese configuration two weight-sharing encoders processing query and document independently produces fixed-length embeddings whose cosine similarity correlates with semantic relatedness. Because document embeddings are computed offline, retrieval at query time reduces to encoding the query and performing a nearest-neighbour lookup over pre-computed vectors. This makes the dense component fast enough to run alongside BM25, and motivates the hybrid first stage adopted in our pipeline.

3 Search Engine Design

3.1 Aim

The aim of this work is to empirically compare two strategies for mitigating the first-stage vocabulary mismatch bottleneck in cascade IR architectures in domain-specific information retrieval. In specialised corpora, queries and documents frequently use different terminology for the same concepts. This problem is particularly acute in financial text, where users pose questions in everyday language while documents employ technical financial vocabulary.

Specifically, we investigate whether a neural approach augmenting the retrieval system with a dense bi-encoder alongside BM25, fused via Reciprocal Rank Fusion or an enriched lexical approach expanding the query through Bo1 pseudo-relevance feedback before BM25 retrieval is more effective at raising first-stage recall and, consequently, end-to-end ranking quality when combined with a cross-encoder reranker. All experiments are conducted on FiQA, where the gap between everyday user queries and technical financial terminology makes the bottleneck both pronounced and representative of broader challenges in domain-specific retrieval.

3.2 Dataset

We select FiQA [18] as our evaluation corpus: a financial question-answering dataset comprising 57,638 passages drawn from StackExchange answers under the Investment topic, paired with 648 test queries and graded relevance judgements. The choice is motivated by three properties that make FiQA an ideal testbed for studying the vocabulary mismatch bottleneck.

First, financial text exhibits particularly severe vocabulary mismatch at two levels: (1) technical synonymy is pervasive *equity valuation* and *stock pricing*, *leveraged buyout* and *LBO* share no surface tokens but denote the same concepts; (2) more critically, there is a systematic register gap between queries and documents: users formulate questions in everyday language (*"how do I save for retirement?"*), while the corpus, written by financially literate StackExchange authors, responds in technical terminology (*"tax-advantaged contribution vehicles"*, *"defined benefit plans"*). This makes FiQA a demanding test case for any lexical first-stage retriever.

Second, FiQA is included in the BEIR benchmark [17], where BM25 achieves one of the lowest nDCG@10 scores (0.236), confirming that the vocabulary mismatch ceiling is not theoretical but measurable. Published baselines also allow direct comparison with our results.

Third, the corpus clearly requires cascade architecture: it is large enough that exhaustive cross-encoder scoring across all 648 queries would require approximately 200 hours on CPU, justifying the cascade design, yet small enough to permit rapid iteration on limited hardware (RTX 4060, 8GB VRAM).

3.3 System Architecture (including Models and Tools)

Our proposed system structure is illustrated in Figure 1. Both pipelines share a two-stage cascade design: a fast first stage retrieves the top 100 candidate documents from the corpus, followed by a cross-encoder reranker that reorders these candidates by relevance. Pipeline A and Pipeline B differ only at Stage 1, enabling a direct comparison between a neural approach (hybrid retrieval) and an enriched lexical approach (query expansion) for addressing vocabulary mismatch at the first retrieval stage.

We hypothesise that each strategy will have its own blindspots. The bi-encoder in Pipeline A captures semantic similarity but was trained on MS MARCO, and whether its representations transfer accurately to financial terminology is uncertain. The query expander in Pipeline B remains within the lexical paradigm and avoids this out-of-domain risk, but cannot match documents that use entirely different vocabulary from the expanded query, as it lacks the ability to understand semantic context. This distinction extending the retriever versus extending the query – defines the central experimental comparison.

Both the Hybrid Retrieval and Query Expander methods aim to surpass the baseline performance of BM25, achieving high recall without relying on computationally expensive methods such as ColBERT.

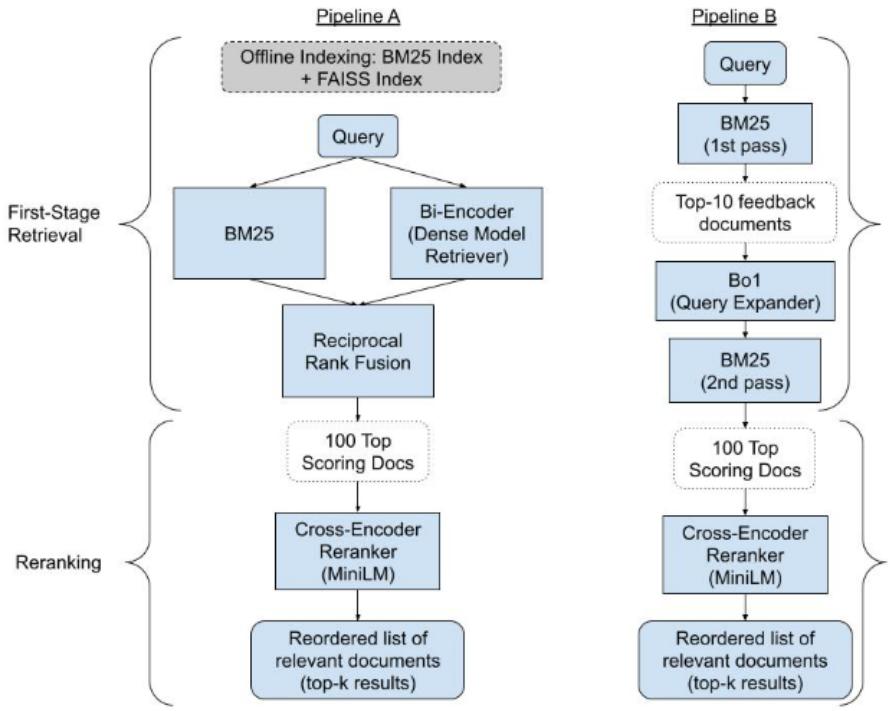


Figure 1: Proposed dual-pipeline architecture. Pipeline A fuses BM25 with a bi-encoder dense retriever via Reciprocal Rank Fusion before BERT reranking. Pipeline B uses Bo1 query expansion before BM25 retrieval and BERT reranking. Both pipelines retrieve the top-100 candidates at the first stage.

3.4 Stage 1: First-Stage Retrieval

The Goal of Stage 1

The goal of Stage 1 is to scan the entire FiQA corpus and return a candidate set of 100 documents per query. This stage serves as a "first-stage filter", because classical lexical retrievers such as BM25 satisfy the speed requirement: inverted index lookup operates in milliseconds over large collections, which is vital for large datasets. The design priority is recall: any relevant document excluded at this stage cannot be recovered by the reranker in Stage 2 regardless of its quality.

3.4.1 Shared Foundation: BM25

BM25 [3] serves as the lexical backbone of both pipelines because it directly satisfies the constraints of Stage 1 defined above. First, speed: BM25 retrieves via inverted index lookup in milliseconds over the entire FiQA corpus (57,638 documents per query) without computational bottleneck. Second, recall: on the BEIR benchmark, BM25 consistently produces high top-100 recall across diverse domains [17], ensuring that the reranker in Stage 2 receives a candidate set containing the majority of relevant documents. Third, reproducibility: BM25 is the standard first-stage baseline in BEIR, and adopting it enables direct comparison of our results with published baselines.

BM25 is implemented via PyTerrier [**?**], chosen for its native integration with ir-datasets and built-in evaluation pipelines. Parameters: $k_1 = 1.2$, $b = 0.75$.

3.4.2 First-Stage Strategy: Two Competing Approaches

Both strategies address the same problem: the vocabulary mismatch ceiling imposed by pure BM25 retrieval (Section 2.1). They differ in how this ceiling is raised. Pipeline A augments the retrieval system itself, adding a dense bi-encoder that captures semantic similarity alongside BM25's lexical matching. Pipeline B instead augments the query, enriching it with related terms so that the existing lexical retriever can match documents it would otherwise miss. This distinction extending the retriever versus extending the query defines the central experimental comparison of the system.

3.4.3 Pipeline A: Hybrid Retrieval

BM25 and a dense bi-encoder retrieve candidates independently and in parallel: BM25 scores documents by term overlap via inverted index, the bi-encoder scores by semantic similarity via nearest-neighbour search over pre-computed embeddings. Each retriever returns a ranked list of 100 candidates. The two lists are then merged using a rank-based fusion method into a single candidate set of up to 100 documents passed to Stage 2. The rationale is complementarity: BM25 reliably retrieves documents that share exact terms with the query financial identifiers, named entities, abbreviations while the bi-encoder retrieves documents that are semantically relevant but use different vocabulary entirely. Neither retriever alone covers both cases.

Dense retrieval model. The bi-encoder model we will use is `all-MiniLM-L6-v2` [15], selected for its efficiency-to-quality ratio: it produces 384-dimensional embeddings, requires only 6 transformer layers, and runs on CPU in under 10ms per query critical properties for a first-stage component that must operate at low latency. Document embeddings are pre-computed offline and stored in a FAISS index [**?**] for nearest-neighbour retrieval at query time.

Rejected alternatives:

- `all-mpnet-base-v2`: 768 dimensions and 12 layers; doubles embedding size and inference cost without proportional recall improvements at top-100.
- DPR [11]: requires GPU-accelerated indexing; shows higher performance degradation on out-of-domain data compared to lighter bi-encoders [17].
- ColBERTv2 [13]: provides richer token-level interactions but requires specialised indexing infrastructure; performance degrades significantly on domain-specific datasets [17].

Fusion method. The two ranked lists will be merged using Reciprocal Rank Fusion [16]: each document receives a score of $\frac{1}{60+r}$ per list, summed across lists. RRF operates on ranks, not raw scores, avoiding the problem of combining BM25 scores and cosine similarities on incomparable scales. It requires no training and no tuning beyond the single constant $k = 60$.

Rejected alternative:

- Linear interpolation ($\alpha \cdot \text{BM25} + \beta \cdot \text{dense}$): requires tuning the mixing weight on held-out data, adding experimental overhead without consistent benefit [16].

3.4.4 Pipeline B: Query Expansion

The original query is expanded with additional terms before BM25 retrieval. An initial BM25 pass identifies a small set of presumably relevant documents; informative terms are extracted from them and appended to the query; a second BM25 pass with the expanded query retrieves the final 100 candidates. The strategy addresses vocabulary mismatch at query time rather than at retrieval time. Retrieval remains entirely lexical this is deliberate, as Pipeline B exists to test whether the vocabulary gap can be closed without neural components, providing a clean experimental contrast with Pipeline A.

We will use Bo1 pseudo-relevance feedback [6]. An initial BM25 pass retrieves a feedback set from which terms are scored by their divergence from a random Bose-Einstein baseline: terms appearing in feedback documents significantly more than their corpus-wide frequency predicts are considered informative and appended to the query. Bo1 is the strongest expansion method that remains entirely within the lexical paradigm. It requires no external model, no GPU, and is implemented natively in PyTerrier, using the same index as Pipeline A.

Rejected alternatives:

- WordNet synonym expansion: general-purpose thesaurus; does not cover financial terminology (e.g., *stock* maps to *livestock* rather than *equity*).
- LLM-based query expansion: semantically strong but introduces query-time inference latency and API dependency, undermining the speed requirement of Stage 1.
- DocT5Query [10]: T5 generates synthetic queries per document at index time; rejected due to T5 inference cost over the full FiQA corpus and the same lexical ceiling limitation.

3.5 Stage 2: Cross-Encoder Reranker

The Goal of Stage 2

Stage 1 prioritises recall, retrieving as many relevant documents as possible while tolerating noise. The resulting candidate set of 100 documents contains relevant results but in an imprecise order: BM25 ranks by term overlap and the bi-encoder by embedding similarity, neither of which captures full cross-attention between query and document tokens. The goal of Stage 2 is to apply a more computationally expensive model that reads query and document jointly to produce a precise relevance score, reordering the candidate set.

A cross-encoder [10] feeds the concatenated sequence `[CLS] query [SEP] document [SEP]` into a BERT-based transformer; the `[CLS]` representation is passed through a linear layer to produce a single relevance score. Unlike the bi-encoder in Stage 1, the cross-encoder processes query and document together, allowing it to capture fine-grained term interactions and context-dependent meaning. One forward pass per query-document pair means that scoring the entire FiQA corpus would require 57,638 forward passes per query – but applied to 100 candidates this is entirely feasible.

The specific model is `cross-encoder/ms-marco-MiniLM-L6-v2`, selected for its inference speed on CPU while maintaining strong reranking quality, and for architectural consistency with the bi-encoder in Stage 1: both use the MiniLM-L6 distilled architecture, keeping the entire system lightweight throughout. The model is loaded via HuggingFace Transformers [**?**].

Rejected alternatives:

- Larger cross-encoders (e.g., `ms-marco-electra-base`): higher quality but approximately 4× slower inference; the marginal gain does not justify the latency across 648 FiQA queries.
- LLM-based reranking [**?**]: setwise zero-shot reranking achieves comparable quality to cross-encoders but is substantially slower due to autoregressive generation, and is competitive with, not better than, cross-encoders.
- MonoT5: T5-based pointwise reranker; competitive quality but higher memory footprint with no advantage over the cross-encoder at the 100-document scale.

3.6 Evaluation Strategy

We evaluate performance at two levels: Stage 1 (first-stage retrieval) and end-to-end (after cross-encoder reranking). This two-level evaluation is essential because our central research question concerns the first stage, but measuring only end-to-end performance would obscure whether improvements originate from better candidate generation or better reranking.

At Stage 1, we use Recall@100 to compare four first-stage configurations: BM25 alone (lexical baseline), Sentence-BERT alone (dense baseline), hybrid retrieval combining both via RRF, and BM25 with Bo1 query expansion. Recall@100 measures what fraction of all relevant documents a strategy captures in its top-100 candidate set – the decisive metric for the first stage, since any relevant document absent from the candidate set is irrecoverable regardless of reranker quality. Importantly, we evaluate BM25 and Sentence-BERT not only in combination but also in isolation to determine whether hybrid improvement stems from the complementarity of lexical and semantic retrieval, or whether the dense component alone is simply a stronger retriever. If Sentence-BERT alone outperforms BM25, the case for fusion weakens; if the hybrid outperforms both, this confirms complementarity.

At the end-to-end level, we report three metrics. **nDCG@10** evaluates the quality of the final top-10 ranking by assigning higher credit to relevant documents placed at earlier positions. This is our primary metric, chosen because it is the BEIR benchmark standard [17], enabling direct comparison with published baselines. **MAP** evaluates precision across all relevant documents in the ranking; FiQA queries typically have multiple relevant documents, and MAP captures how consistently a system retrieves all of them, complementing nDCG@10's focus on the top of the ranking. **MRR@10** captures the position of the first relevant document in the top-10, reflecting the user experience of how many results must be examined before finding a relevant answer.

To isolate the contribution of each component, we compare the configurations summarised in Table **??**.

Table 1: Retrieval configurations and evaluation metrics used in the experiments.

Configuration	Stage 1	Stage 2	Tests	Recall@100	nDCG@10	MAP	MRR@10
BM25	Lexical	–	Lexical baseline	✓	✓	✓	✓
SBERT	Dense	–	Dense baseline	✓	✓	✓	✓
Hybrid	Hybrid	–	Value of fusion	✓	✓	✓	✓
BM25 + Bo1	Expanded lexical	–	Value of expansion	✓	✓	✓	✓
BM25 + CE	Lexical	Cross-encoder	Reranking on lexical	–	✓	✓	✓
SBERT + CE	Dense	Cross-encoder	Reranking on dense	–	✓	✓	✓
Hybrid + CE	Hybrid	Cross-encoder	Pipeline A	–	✓	✓	✓
BM25+Bo1+CE	Expanded lexical	Cross-encoder	Pipeline B	–	✓	✓	✓

All metrics are computed using `pytrec_eval` [19]. Relevance judgements come directly from the FiQA benchmark, collected via a Cranfield-style pooled evaluation in which assessors manually scored document relevance from pooled candidate sets.

4 Workplan and Team Responsibilities

4.1 Tasks

The project addresses the problem of designing a search engine that can reliably retrieve and rank relevant passages for user questions in the specialised financial corpus FiQA, while mitigating vocabulary mismatch between everyday language and domain-specific financial terminology. We decompose this into four tasks:

1. **Baseline retrieval.** Build a BM25 first-stage retriever over FiQA and establish baseline retrieval performance as a reference point for all subsequent experiments.
2. **Neural reranking.** Apply a BERT-based cross-encoder to rerank the BM25 candidates and quantify improvements in top-k ranking quality on FiQA.
3. **First-stage comparison.** Implement and compare two strategies for raising the Stage 1 recall ceiling: (a) hybrid retrieval combining BM25 with a Sentence-BERT bi-encoder via Reciprocal Rank Fusion, and (b) Bo1 pseudo-relevance feedback query expansion, testing whether extending the retriever or extending the query better mitigates vocabulary mismatch.

4. **Evaluation and analysis.** Evaluate all configurations on FiQA using standard IR metrics and analyse which setup yields the best passage ranking performance for financial question answering.

4.2 Team Responsibilities

Table 2: Team member responsibilities derived from project task breakdown

Member	Primary Role	Tasks
Edwards, Samuel	Dense Retrieval & Fusion	Dense retriever implementation, FAISS indexing, RRF fusion integra
Mathur, Hardik	Evaluation & Testing	Evaluation setup, test runs, metric computation via pytrec_eval
Sheikh, Rayaan	Neural Reranking	BERT cross-encoder integration, Zero-shot integration
Zeleniuk, Artem	BM25 & Query Expansion	BM25 indexing and retrieval setup, query expansion module (Bo1)

All members share responsibility for writing and reviewing the final report.

4.3 Timeline

The development is organised into parallel workstreams reflecting the modular architecture of the system. Core retrieval components BM25 indexing, the dense retriever with FAISS indexing, and the query expansion module are built in parallel during the first two weeks. RRF fusion is integrated once both retrievers are operational. The BERT cross-encoder and an LLM zero-shot module are added in the later stages. The final phase focuses on evaluation setup and test runs.

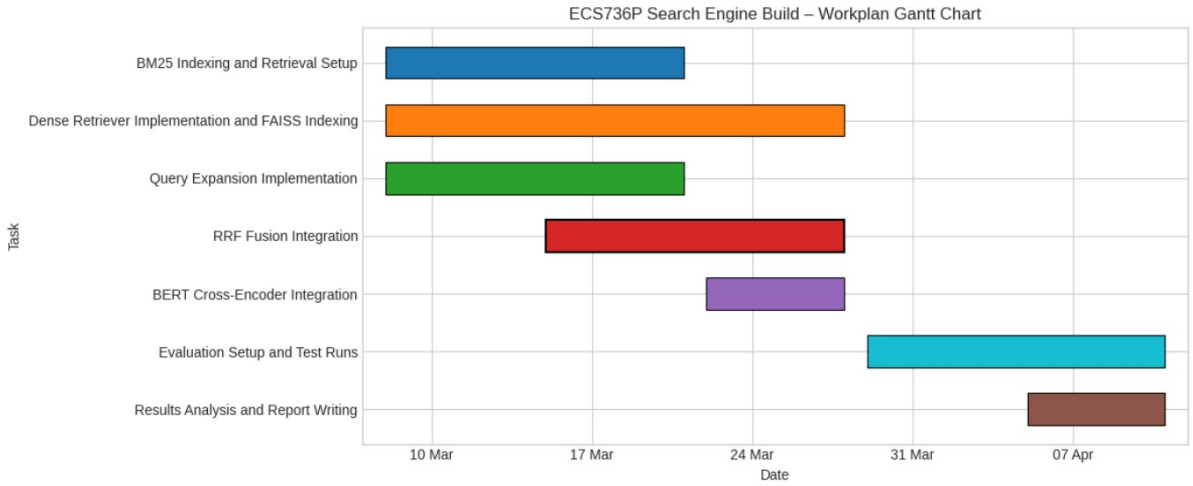


Figure 2: Project Gantt chart covering early March to early April. Retrieval components are built in parallel before integration and final evaluation.

5 Conclusion

In conclusion, to address the vocabulary mismatch found within the financial domain and FiQA dataset, we propose a two-stage cascade architecture. This will compare a lexical query expansion approach with a dense hybrid approach, followed by BERT reranking. We hypothesise that the encoders will have a contextual blindspot when used on our domain. They have been trained on the MS MARCO dataset, which is generic; as a result they will likely perform much worse on a domain-specific dataset such as FiQA. An approach which does not imply understanding of context but relies on query expansion may perform better.

Future work could replace BM25 with a dense or late-interaction first-stage retriever such as ColBERT, which would address the vocabulary mismatch bottleneck more directly. If we were to bring the higher-scoring system to production, we would employ a front-end system such as Elasticsearch to manage user queries at scale.

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